GUIDE 4:

Optimisation of your

Magic Moving Average Expert Advisor

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Overview

Previous Guides covered -

1. Introduction to the Magic Moving Average Expert Advisor
2. Ensuring you have reasonably good Test Data,
3. Initial Setting up and testing of the Magic Moving Average Expert Advisor

This fourth guide will cover the fun and important part and that is finding the most suitable settings for your Magic Moving Average expert Advisor. We are going to start where guide 3 stopped after the Strategy Tester was set up and we did the initial Magic Moving Average test using the initial settings.

This Fourth Guide will cover

- General optimisation considerations
- Optimisation using many variables
- Refining the results using Heat maps
- Loading your settings into the Expert Advisor

It is important to note that we are not trying to teach you how to become professional and perfect EA optimisers. This can take a long time and requires considerable experience. We want you to know enough about the optimisation process for you to find workable settings for the Magic Moving Average Expert Advisor.

In Guide 3 we saw that the initial test of the Magic Moving Average has produced good results but not that exciting. 3842 pips in 24 months is not that great so let's see if we can improve on that.

When we optimise settings we are asking the Strategy tester to test thousands of possible combinations of settings. The greater the number of combinations the more time the process will take – sometimes this process can take many, many hours or even days.

The main factors impacting the speed of optimisation are

- The method used – tick or open prices
- The timeframe selected – 4 hour time frame will be faster than the 5 minute one
- The period used for optimisation – 12 years will take longer than 2 years
- The number of combinations tested – 96 combinations will be much faster than 24,000 combinations
- The capacity of your computer – always make sure your temporary memory is cleared and your computer has been rebooted recently
- Sometimes the complexity of your EA

- Whether the Genetic Algorithm was selected for optimisation

It is therefore important to focus on the main variables that are going to have the biggest financial impact on your results.

For purposes of this initial test of the Magic Moving Average Expert Advisor we are going to focus on

- the time of day hours FROM variable;
- the time of day hours TO variable;
- the Moving Average Period;
- the Moving Average offsets.

So let's start:-

Determining the range and optimisation steps

For the Time of day variable we want to cover the whole day. This would mean that we would start at 0 hour and end a 23 hour as our FROM hours variable. Our TO hours variable will than start at 1 and stop at 24. This will cover the whole day in terms of the Hours variable.

At this stage we will not refine these any further by introducing optimisation of the Minute FROM and TO variables. This can be done as further refinement at a later stage.

For the Moving Average Period variable we have chosen any setting between 5 and 200 with steps of 5. The reason is that we do not want settings over 200 for the MA period. If you prefer period even lower that 200 then adjust this value to your personal preference.

In terms of offset we have selected offsets from 2 to 60 with small gaps of 2. This gives 30 alternatives. Again if you have personal preferences use those.

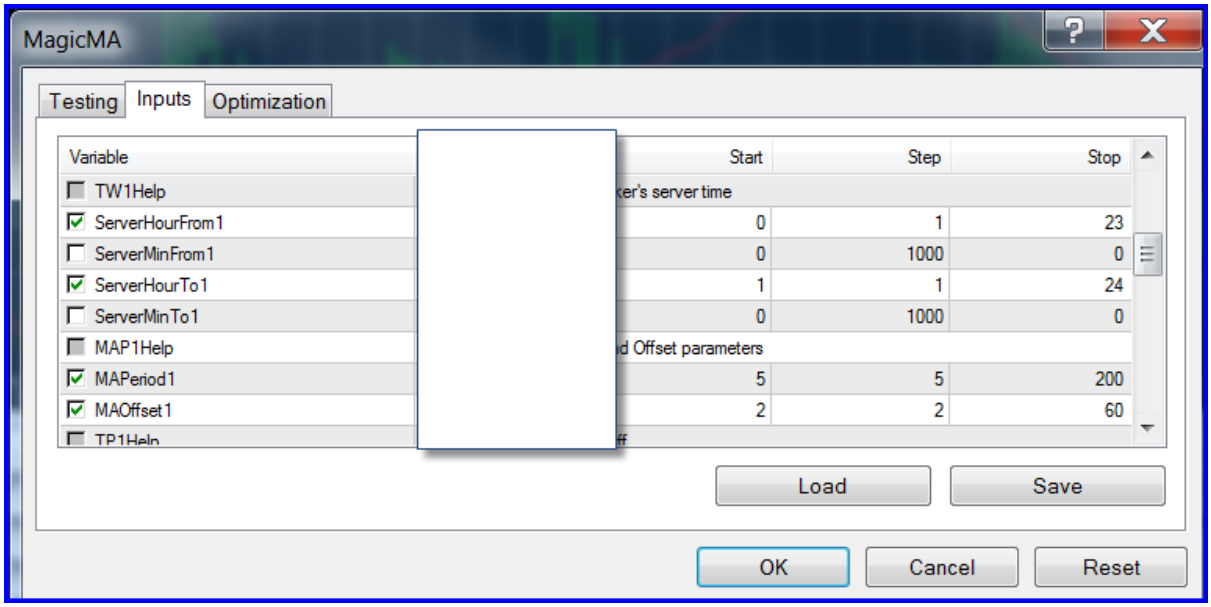
Some traders have 0 as an offset setting and would then not optimise this variable. The result would then be a pure moving average period amount. We have found the period setting increase tremendously when this is done.

The Table below reflects the optimisation test levels for the Start, Step and Stop for the Volume, Target and Stop variables. The number of setting combinations using these alternatives amounts to 691 200 (24 x 24 x 40 x 30) during these optimisation runs.

Optimisation settings	Start	Step	Stop	Alternatives
ServerHoursFrom1	0	1	23	24
ServerHoursTo1	1	1	24	24
MAPeriod1	5	5	200	40
MAoffset1	2	2	60	30

Testing 691,200 combinations

The setting would then look like the picture below. Note that only the variables we are going to optimise are the ones ticked. No other variable is ticked.



Confirming the EA setting used to optimise the EA

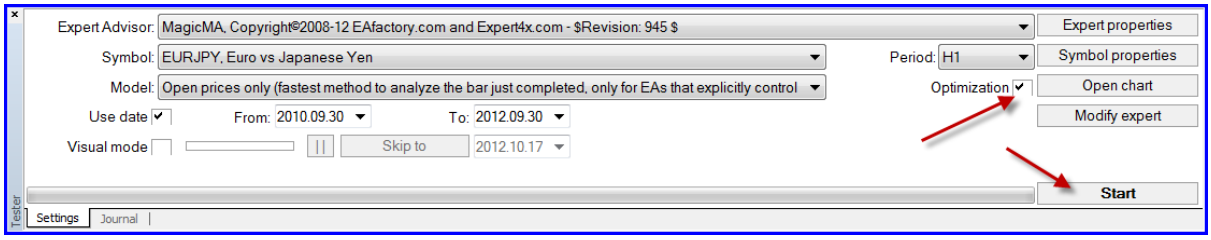
Before we run our optimisation test let's confirm the overall settings that will be used are the ones below. They are the same ones used to generate 3 842 pips in 24 months in Guide 3

Variable	Value	Start	Step	Stop
DaysHelp	Allow trade for a day 0:Off 1:Active			
Monday	1	1	0	0
Tuesday	1	1	0	0
Wednesday	1	1	0	0
Thursday	1	1	0	0
Friday	1	1	0	0
Saturday	0	1	0	0
Sunday	0	1	0	0

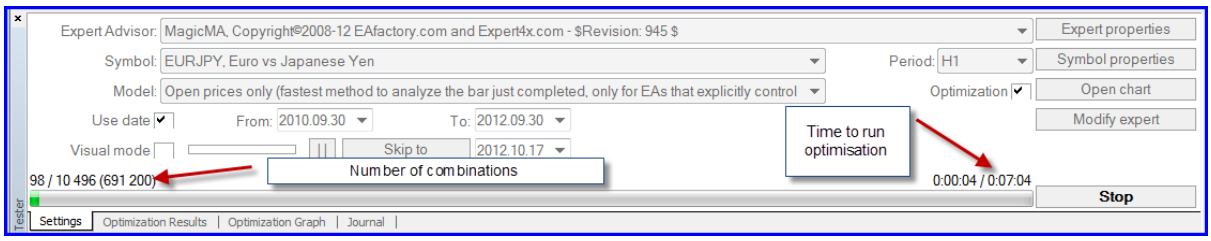
Variable	Value	Start	Step	Stop
Lots	0.1	0.1	0	0
TW1Help	First trading window in broker's server time			
ServerHourFrom1	0	0	1	23
ServerMinFrom1	0	0	0	0
ServerHourTo1	24	1	1	24
ServerMinTo1	0	0	1000	5000
MAP1Help	Moving Average Period and Offset parameters			
MAPeriod1	65	5	5	200
MAOffset1	40	2	2	60

Variable	Value	Start	Step	Stop
MAPeriod1	65	5	5	200
MAOffset1	40	2	2	60
TP1Help	Set target profit in pips 0:Off			
TakeProfit1	500	50	50	800
TS1Help	Set trailing stop in pips 0:Off			
TrailingStop1	0	50	50	800
SWEHelpHelp	Activate 2nd trading window 0:Off 1:Active			
SecondWindowEnabled	0	0	0	0

The next step is to merely tick the optimisation block and click on the start button



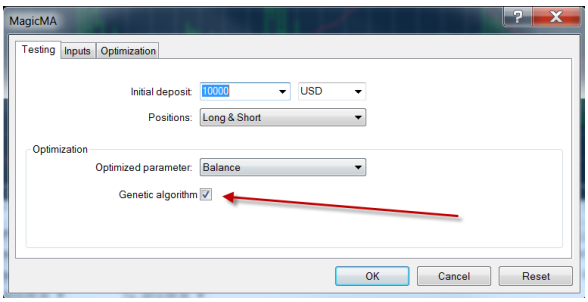
The Strategy tester will then start the optimisation process. As it starts it will give you an idea of how long it will take. As seen below it will take just under 7 minutes. It also shows that it is dealing with 691 200 combinations. Not bad for a simple laptop.



There are a number of factors impacting your testing and optimisation speeds. Much of the processing done for the course was done on a 3 year old laptop. Computer speed does play a role but not a major one.

Firstly: You need to be testing using the **"Open Prices Only"** option when using the Strategy testing. To use this option make sure that your EA makes major decisions on the close of the candle. This will increase speed considerably

Secondly: You need to be using the **Genetic Algorithm** by having this option ticked in the "Testing" tab of your EA properties. This will make a major impact on processing speeds.

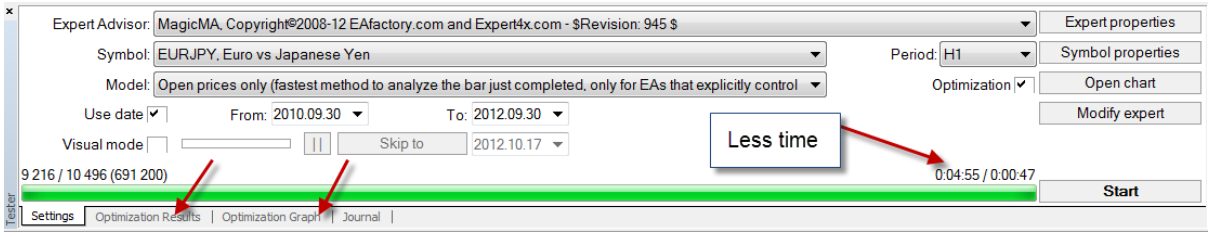


Thirdly: Your computer memory (RAM) should be a clean as possible. To do this, merely delete all your temporary internet files via your browser or by using the "Disk Cleanup" function. Then reboot your computer before testing and optimising. This will speed your computers processing speed up.

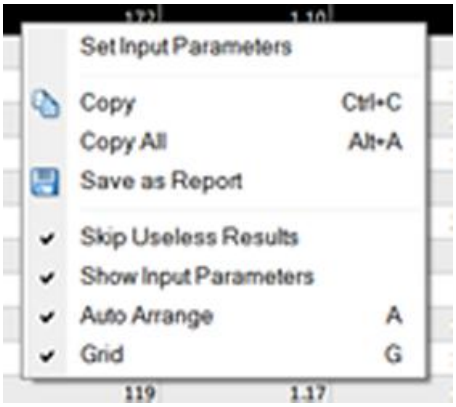
Optimised Results

So after a few minutes these are the results. The Strategy Tester presents its results in 2 formats.

1. An Optimisation graph
2. Optimisation results in table form.



We are most interested in the results in table form at this stage. The table format only shows the positive results as a default. There are many options you can specify for your results table. You can specify the options by right clicking on the optimised results page. You can also sort most columns in this table. The table that follows has been sorted in order of high profit to low profit.



The final results of the optimisation run sorted by profit

Pass	Profit	Total trades	Profit factor	Expected Payoff	Drawdown $	Drawdown %
125	4661.44	140	2.56	33.30	835.72	8.21
2381	4629.07	326	1.59	14.20	1590.02	15.46
2610	4624.89	215	1.81	21.51	851.55	6.89
3352	4621.47	316	1.61	14.62	935.30	8.69
2183	4608.77	162	2.24	28.45	930.13	8.89
2720	4607.12	361	1.51	12.76	1415.29	13.74
2114	4595.03	461	1.41	9.97	2021.05	19.32
816	4578.18	154	2.43	29.73	886.93	8.72
3385	4557.41	358	1.52	12.73	1052.79	9.45
2355	4553.58	448	1.42	10.16	1916.09	18.32
2279	4525.79	176	2.15	25.71	946.09	9.08
2751	4503.60	329	1.56	13.69	935.10	8.65
1878	4501.98	288	1.57	15.63	1486.93	14.37
3222	4497.29	307	1.55	14.65	1193.48	11.57
2561	4492.06	471	1.39	9.54	1907.22	18.23
2782	4490.18	224	1.77	20.05	871.13	6.95
2848	4490.14	374	1.47	12.01	1278.16	12.41
2214	4470.15	301	1.56	14.85	1512.11	14.66
3355	4450.56	458	1.40	9.72	1802.25	17.23
3154	4449.53	270	1.68	16.48	841.84	8.24
2967	4422.87	261	1.71	16.95	781.21	7.63
3001	4399.37	170	2.11	25.88	1068.00	8.38
2027	4392.17	300	1.59	14.64	978.75	9.58
2416	4387.28	297	1.57	14.77	1353.56	13.02
2261	4383.90	307	1.53	14.28	985.91	8.22

Overall result

The initial reaction to the optimisation run is that we have increased the profit since our initial setting test considerably to $4 661. Also you can see that there are literally hundreds of profitable settings.

The top result is not always the one we are going to grab for live trading. There are a few other factors to consider.

Sorting by low drawdown

We are looking for settings that give a **low drawdown** figure. The Drawdown is an indication of risk and stability. If the drawdown is low, the risk and stability of the system is good. So, sorting the results by drawdown percentage results in the table that follows:

Pass	Profit	Total trades	Profit factor	Expected Payoff	Drawdown $	Drawdown...	Inputs
404	588.31	11	5.59	53.48	160.93	1.59	ServerHourFrom1=23; ServerHourTo1=24; MAPeriod1=80; MAOffset1=58; ...
2227	1158.18	8	11.29	144.77	197.69	1.86	ServerHourFrom1=23; ServerHourTo1=24; MAPeriod1=80; MAOffset1=34; ...
1874	26.68	11	1.15	2.43	191.17	1.90	ServerHourFrom1=7; ServerHourTo1=8; MAPeriod1=40; MAOffset1=60; Mo...
1902	505.33	10	3.71	50.53	200.84	1.98	ServerHourFrom1=23; ServerHourTo1=24; MAPeriod1=160; MAOffset1=34; ...
392	87.39	15	1.32	5.83	206.17	2.04	ServerHourFrom1=7; ServerHourTo1=8; MAPeriod1=40; MAOffset1=32; Mo...
58	1107.51	10	7.81	110.75	206.80	2.07	ServerHourFrom1=23; ServerHourTo1=24; MAPeriod1=135; MAOffset1=20; ...
72	1109.34	14	6.16	79.24	236.68	2.09	ServerHourFrom1=5; ServerHourTo1=6; MAPeriod1=75; MAOffset1=32; Mo...
3274	316.34	13	2.23	24.33	220.96	2.21	ServerHourFrom1=1; ServerHourTo1=2; MAPeriod1=75; MAOffset1=30; Mo...
628	250.40	13	2.05	19.26	237.66	2.27	ServerHourFrom1=0; ServerHourTo1=1; MAPeriod1=165; MAOffset1=34; M...
2911	11.32	7	1.14	1.62	255.31	2.49	ServerHourFrom1=23; ServerHourTo1=24; MAPeriod1=200; MAOffset1=48; ...
13	472.58	16	2.77	29.54	253.02	2.51	ServerHourFrom1=23; ServerHourTo1=24; MAPeriod1=105; MAOffset1=2; ...
90	396.41	12	2.66	33.03	256.74	2.55	ServerHourFrom1=5; ServerHourTo1=6; MAPeriod1=195; MAOffset1=22; M...
3369	764.88	33	2.60	23.18	259.58	2.56	ServerHourFrom1=1; ServerHourTo1=3; MAPeriod1=105; MAOffset1=30; M...
203	75.99	29	1.17	2.62	261.40	2.61	ServerHourFrom1=21; ServerHourTo1=22; MAPeriod1=20; MAOffset1=14; ...
2955	212.27	10	1.71	21.23	282.28	2.79	ServerHourFrom1=1; ServerHourTo1=2; MAPeriod1=75; MAOffset1=36; Mo...
705	588.01	21	2.24	28.00	284.92	2.80	ServerHourFrom1=11; ServerHourTo1=12; MAPeriod1=75; MAOffset1=46; ...
200	1518.07	23	4.47	66.00	321.77	2.86	ServerHourFrom1=5; ServerHourTo1=7; MAPeriod1=55; MAOffset1=44; Mo...
2443	10.49	18	1.02	0.58	288.76	2.87	ServerHourFrom1=1; ServerHourTo1=2; MAPeriod1=30; MAOffset1=50; Mo...
2774	300.69	11	3.06	27.34	301.48	2.93	ServerHourFrom1=20; ServerHourTo1=21; MAPeriod1=115; MAOffset1=12; ...
3271	495.73	26	1.98	19.07	315.12	2.96	ServerHourFrom1=0; ServerHourTo1=1; MAPeriod1=5; MAOffset1=36; Mon...
95	695.12	14	3.26	49.65	318.31	2.96	ServerHourFrom1=21; ServerHourTo1=22; MAPeriod1=100; MAOffset1=58; ...
3242	2613.04	23	7.98	113.61	319.12	3.00	ServerHourFrom1=16; ServerHourTo1=17; MAPeriod1=40; MAOffset1=58; ...
187	251.47	21	1.63	11.97	325.05	3.07	ServerHourFrom1=21; ServerHourTo1=22; MAPeriod1=100; MAOffset1=10; ...
721	1012.11	27	3.08	37.49	327.79	3.08	ServerHourFrom1=6; ServerHourTo1=8; MAPeriod1=95; MAOffset1=4; Mon...
41	491.96	14	2.35	35.14	334.53	3.09	ServerHourFrom1=21; ServerHourTo1=22; MAPeriod1=115; MAOffset1=48; ...
1018	66.73	12	1.25	5.56	336.66	3.34	ServerHourFrom1=20; ServerHourTo1=21; MAPeriod1=185; MAOffset1=8;

This has not helped much as there are no real profitable combinations with a low drawdown percentage. We can ignore the option of using a low drawdown to earn high returns.

Sorting by Profit factor

The other factor showing risk in a technique is the **profit factor**. This is basically the ratio of the value of profitable trades to the value of negative trades. We want this to be high and over 1.5 to 2.0 would be good. So we can sort the table using the profit factor.

Pass	Profit	Total trades	Profit factor	Expected Payoff	Drawdown $	Drawdown %	Inputs
2227	1158.18	8	11.29	144.77	197.69	1.86	ServerHourFrom1=23; ServerHourTo1=24; MAPeriod1=80; MAOffset1=34; ...
3242	2613.04	23	7.98	113.61	319.12	3.00	ServerHourFrom1=16; ServerHourTo1=17; MAPeriod1=40; MAOffset1=58; ...
58	1107.51	10	7.81	110.75	206.80	2.07	ServerHourFrom1=23; ServerHourTo1=24; MAPeriod1=135; MAOffset1=20; ...
2867	2752.99	28	6.80	98.32	391.38	3.65	ServerHourFrom1=16; ServerHourTo1=17; MAPeriod1=40; MAOffset1=60; ...
72	1109.34	14	6.16	79.24	236.68	2.09	ServerHourFrom1=5; ServerHourTo1=6; MAPeriod1=75; MAOffset1=32; Mo...
404	588.31	11	5.59	53.48	160.93	1.59	ServerHourFrom1=23; ServerHourTo1=24; MAPeriod1=80; MAOffset1=58; ...
200	1518.07	23	4.47	66.00	321.77	2.86	ServerHourFrom1=5; ServerHourTo1=7; MAPeriod1=55; MAOffset1=44; Mo...
3070	1218.01	20	4.13	60.90	450.62	4.13	ServerHourFrom1=2; ServerHourTo1=3; MAPeriod1=70; MAOffset1=30; Mo...
2941	3462.81	61	4.07	56.77	469.78	4.61	ServerHourFrom1=2; ServerHourTo1=6; MAPeriod1=75; MAOffset1=30; Mo...
2223	2225.39	43	4.05	51.75	467.93	3.94	ServerHourFrom1=3; ServerHourTo1=6; MAPeriod1=75; MAOffset1=32; Mo...
1902	505.33	10	3.71	50.53	200.84	1.98	ServerHourFrom1=23; ServerHourTo1=24; MAPeriod1=160; MAOffset1=34; ...
1831	3208.35	63	3.62	50.93	554.67	5.46	ServerHourFrom1=2; ServerHourTo1=6; MAPeriod1=75; MAOffset1=32; Mo...
213	1199.77	17	3.55	70.57	526.67	4.49	ServerHourFrom1=9; ServerHourTo1=10; MAPeriod1=160; MAOffset1=6; M...
1228	2327.50	43	3.33	54.13	522.52	5.07	ServerHourFrom1=16; ServerHourTo1=18; MAPeriod1=70; MAOffset1=30; ...
95	695.12	14	3.26	49.65	318.31	2.96	ServerHourFrom1=21; ServerHourTo1=22; MAPeriod1=100; MAOffset1=58; ...
2161	2517.48	60	3.24	41.96	648.70	6.35	ServerHourFrom1=1; ServerHourTo1=5; MAPeriod1=75; MAOffset1=30; Mo...
1102	1952.84	46	3.23	42.45	356.54	3.54	ServerHourFrom1=5; ServerHourTo1=8; MAPeriod1=20; MAOffset1=46; Mo...
293	2299.86	46	3.14	50.00	474.97	4.61	ServerHourFrom1=16; ServerHourTo1=18; MAPeriod1=75; MAOffset1=26; ...
3499	1776.67	26	3.14	68.33	361.69	3.44	ServerHourFrom1=16; ServerHourTo1=17; MAPeriod1=70; MAOffset1=34; ...
3325	3957.73	96	3.12	41.23	600.29	5.95	ServerHourFrom1=2; ServerHourTo1=9; MAPeriod1=75; MAOffset1=30; Mo...
721	1012.11	27	3.08	37.49	327.79	3.08	ServerHourFrom1=6; ServerHourTo1=8; MAPeriod1=95; MAOffset1=4; Mon...
994	2845.81	49	3.07	58.08	1034.28	8.15	ServerHourFrom1=16; ServerHourTo1=18; MAPeriod1=90; MAOffset1=58; ...

Sorting by profit factor has not really resulted in very high profit figures.

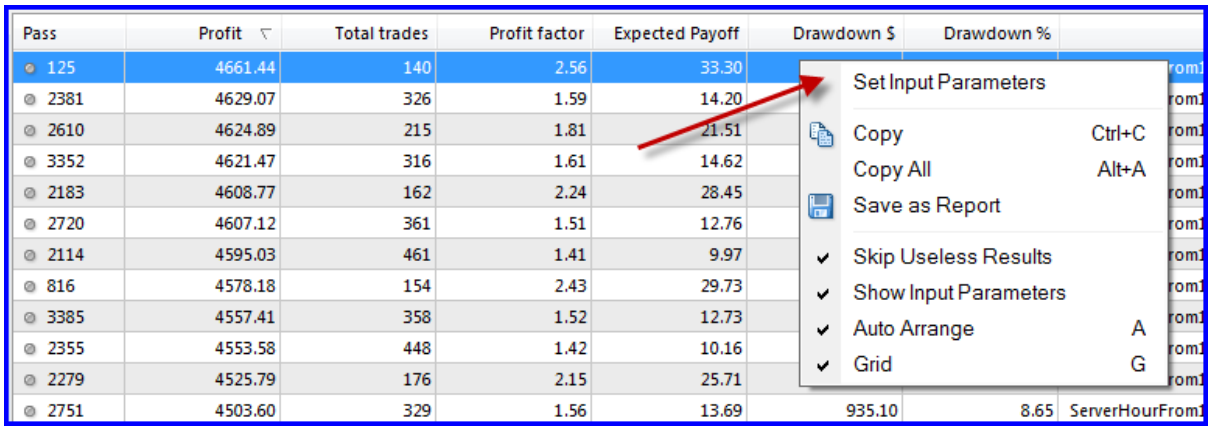
Results are still low?

In general there must be a balance between profits, profitability (Profit factor) and risk (Drawdowns). In the end total profits is be the main objective of most traders.

The next major step to finding possible increase profits would be to change the currency or the time frame and redo this process. This has been done for you in Guide 5 by optimising 18 currencies and 5 timeframes.

Loading optimised settings automatically

If you want to load an optimised setting into the EA settings from the optimisation results, you merely have to right click on the setting you want to load, select “Set Input Parameters” and the optimised settings will be loaded into your EA automatically.



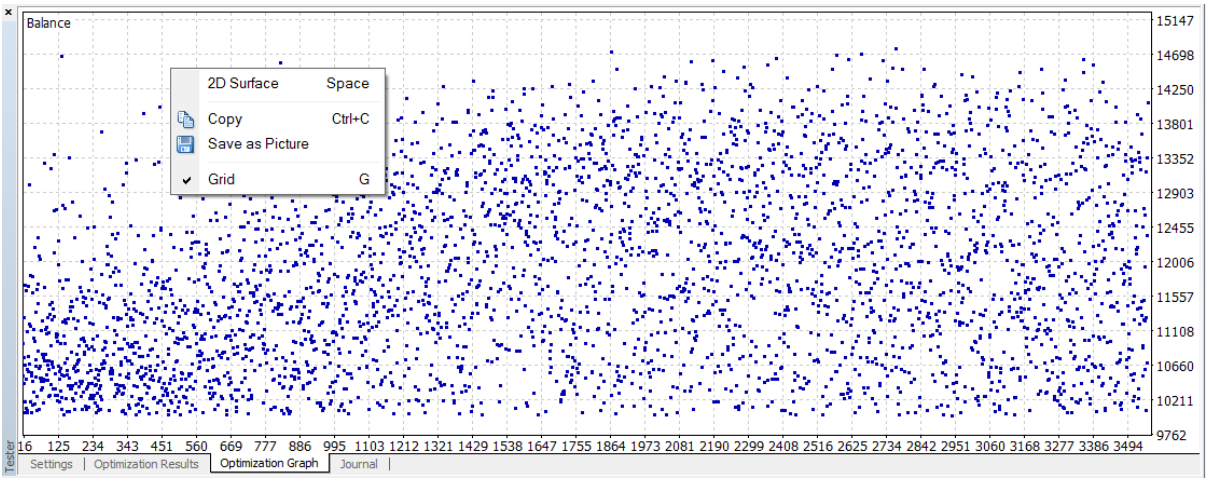
Refining our findings using the "HEAT MAPS" provided by the Strategy Tester

Often optimised results carry more risk. Especially if they are very specific.

Before going live, it would be good to analyse some of the results in more detail. We do this by using the Heat maps provided by the Strategy Tester.

We did not discuss the Optimisation Graphs in great detail. These can supply additional information for deciding on final settings for your EA.

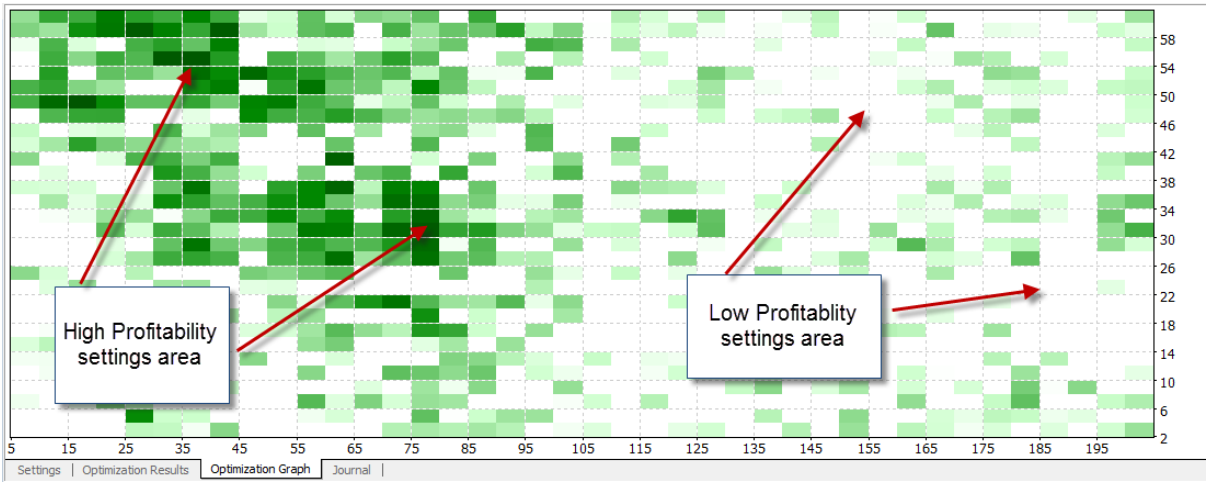
The normal view of the optimisation graph, as shown below, shows the distribution of profitability of the options tested.



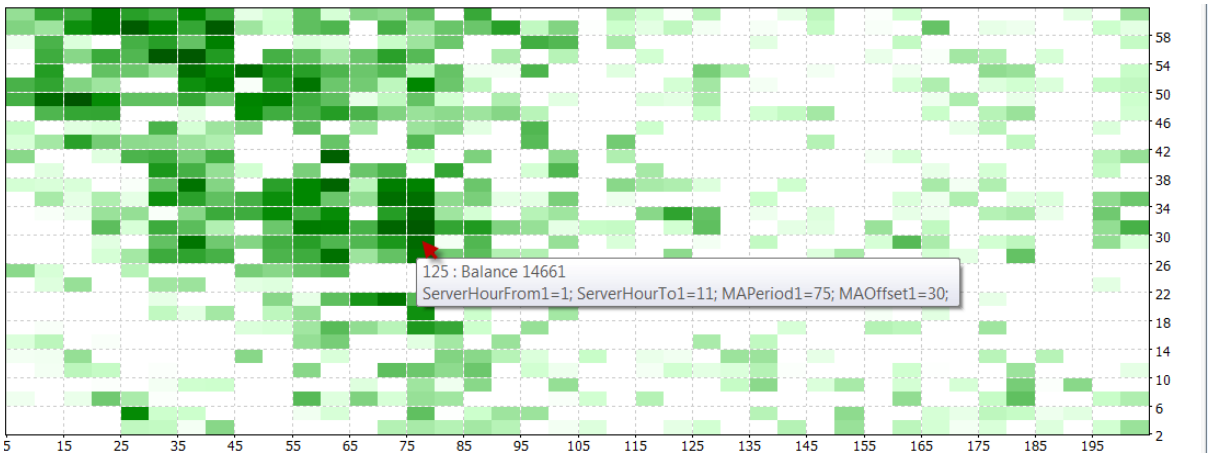
The true value comes from the heat maps that can be created using the graph information. By right clicking in the graph you get a small window with a number of options. See the small window in the picture above.

If you then click on the 2D surface option, you are presented with a heat map showing the relationship between any of the optimised variables that you can place on the x and y axis.

If you right-click on the heat map it gives you options to vary the x axis and the y axis. So you can create heat maps showing where the most profitable settings are located for example, the MA period and MA offsets combinations as shown below. The darker the colour, the more concentrated the profitable combinations.



Further, if you hover your cursor over the darkest block it will show you the most profitable setting combinations in total.

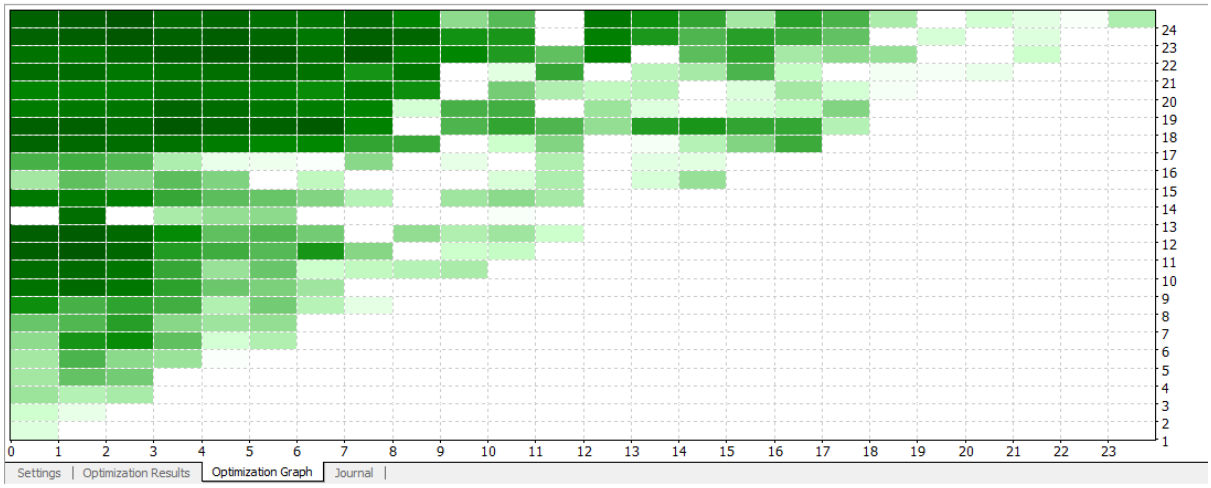


You can then trace the suggested settings by relating them to the Optimization results report using the pass number (125) or by double-clicking the relevant block.

Pass	Profit	Total trades	Profit factor	Expected Payoff	Drawdown $	Drawdown %	Inputs
125	4661.44	140	2.56	33.30	835.72	8.21	ServerHourFrom1=1; ServerHourTo1=11; MAPeriod1=75; MAOffset1=30; ...

The heat map gives a good visual view of where the profitable settings are. It is also important to look at the blocks surrounding the darkest block to see if they are also dark. If they are, it gives you confidence that if the settings are not perfect but they are neighbours to settings that also have a good chance of being profitable.

Doing a heat map using FROM hours as the X-Axis and the TO hours the Y-Axis give an incredibly clear picture of profitable setting and unprofitable settings as seem below. The dark green combinations are much more concentrated and the white blocks are also much more concentrated. Heat maps not only give you areas where the combination are good but also areas where the combinations are very poor.



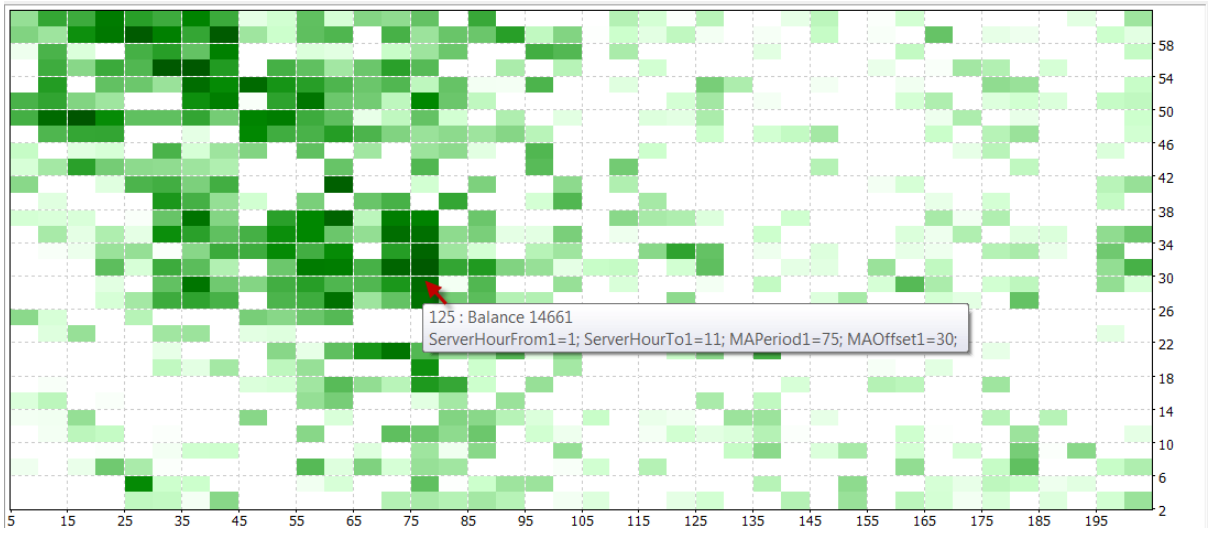
Making a final decision on the settings

From the review of the heat maps we can see that as long as we use setting that are surrounded by dark blocks or that are in the areas where dark green blocks are concentrated the setting should be sound. Looking at the alternative combinations presented below there are no combinations that have a better combination of profit, profit factor and drawdown % that the existing first option.

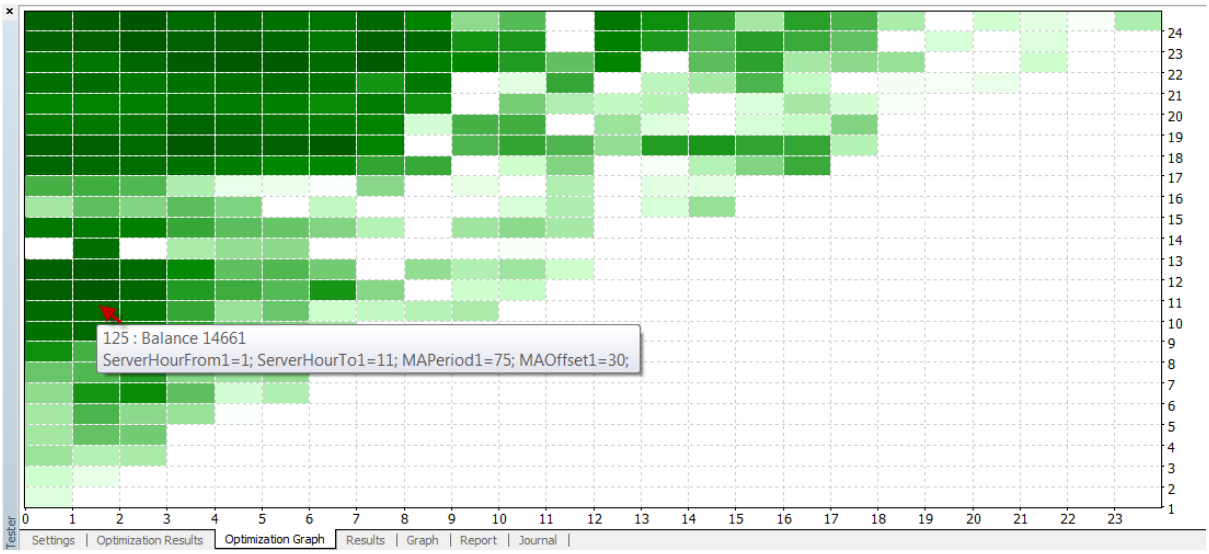
We have already determined that in terms of heat maps the setting fall in the areas with high concentrations of dark green so we can go with Pass 125 as our selection.

Pass	Profit	Total trades	Profit factor	Expected Payoff	Drawdown $	Drawdown %	Inputs
125	4661.44	140	2.56	33.30	835.72	8.21	ServerHourFrom1=1; ServerHourTo1=11; MAPeriod1=75; MAOffset1=30; ...
2381	4629.07	326	1.59	14.20	1590.02	15.46	ServerHourFrom1=5; ServerHourTo1=22; MAPeriod1=25; MAOffset1=58; ...
2610	4624.89	215	1.81	21.51	851.55	6.89	ServerHourFrom1=6; ServerHourTo1=18; MAPeriod1=40; MAOffset1=58; ...
3352	4621.47	316	1.61	14.62	935.30	8.69	ServerHourFrom1=7; ServerHourTo1=22; MAPeriod1=30; MAOffset1=54; ...
2183	4608.77	162	2.24	28.45	930.13	8.89	ServerHourFrom1=1; ServerHourTo1=12; MAPeriod1=75; MAOffset1=30; ...
2720	4607.12	361	1.51	12.76	1415.29	13.74	ServerHourFrom1=3; ServerHourTo1=22; MAPeriod1=30; MAOffset1=54; ...
2114	4595.03	461	1.41	9.97	2021.05	19.32	ServerHourFrom1=1; ServerHourTo1=24; MAPeriod1=15; MAOffset1=48; ...
816	4578.18	154	2.43	29.73	886.93	8.72	ServerHourFrom1=0; ServerHourTo1=11; MAPeriod1=75; MAOffset1=30; ...
3385	4557.41	358	1.52	12.73	1052.79	9.45	ServerHourFrom1=4; ServerHourTo1=23; MAPeriod1=30; MAOffset1=54; ...
2355	4553.58	448	1.42	10.16	1916.09	18.32	ServerHourFrom1=1; ServerHourTo1=23; MAPeriod1=15; MAOffset1=48; ...
2279	4525.79	176	2.15	25.71	946.09	9.08	ServerHourFrom1=0; ServerHourTo1=12; MAPeriod1=75; MAOffset1=30; ...
2751	4503.60	329	1.56	13.69	935.10	8.65	ServerHourFrom1=7; ServerHourTo1=23; MAPeriod1=30; MAOffset1=54; ...
1878	4501.98	288	1.57	15.63	1486.93	14.37	ServerHourFrom1=1; ServerHourTo1=18; MAPeriod1=75; MAOffset1=30; ...
3222	4497.29	307	1.55	14.65	1193.48	11.57	ServerHourFrom1=0; ServerHourTo1=18; MAPeriod1=60; MAOffset1=40; ...
2561	4492.06	471	1.39	9.54	1907.22	18.23	ServerHourFrom1=0; ServerHourTo1=24; MAPeriod1=15; MAOffset1=48; ...
2782	4490.18	224	1.77	20.05	871.13	6.95	ServerHourFrom1=5; ServerHourTo1=18; MAPeriod1=40; MAOffset1=58; ...
2848	4490.14	374	1.47	12.01	1278.16	12.41	ServerHourFrom1=3; ServerHourTo1=23; MAPeriod1=30; MAOffset1=54; ...
2214	4470.15	301	1.56	14.85	1512.11	14.66	ServerHourFrom1=0; ServerHourTo1=18; MAPeriod1=75; MAOffset1=30; ...
3355	4450.56	458	1.40	9.72	1802.25	17.23	ServerHourFrom1=0; ServerHourTo1=23; MAPeriod1=15; MAOffset1=48; ...
3154	4449.53	270	1.68	16.48	841.84	8.24	ServerHourFrom1=4; ServerHourTo1=18; MAPeriod1=35; MAOffset1=54; ...
2967	4422.87	261	1.71	16.95	781.21	7.63	ServerHourFrom1=5; ServerHourTo1=18; MAPeriod1=35; MAOffset1=54; ...
3001	4399.37	170	2.11	25.88	1068.00	8.38	ServerHourFrom1=0; ServerHourTo1=11; MAPeriod1=60; MAOffset1=36; ...
2027	4392.17	300	1.59	14.64	978.75	9.58	ServerHourFrom1=3; ServerHourTo1=19; MAPeriod1=35; MAOffset1=54; ...
2416	4387.28	297	1.57	14.77	1353.56	13.02	ServerHourFrom1=0; ServerHourTo1=17; MAPeriod1=75; MAOffset1=32; ...
2261	4383.90	307	1.53	14.28	985.91	8.22	ServerHourFrom1=1; ServerHourTo1=18; MAPeriod1=70; MAOffset1=30; ...
2678	4370.74	339	1.53	12.89	1642.74	15.97	ServerHourFrom1=5; ServerHourTo1=23; MAPeriod1=25; MAOffset1=58; ...
3294	4360.60	367	1.48	11.88	1052.83	9.45	ServerHourFrom1=4; ServerHourTo1=24; MAPeriod1=30; MAOffset1=54; ...
1803	4345.59	334	1.53	13.01	1654.86	16.09	ServerHourFrom1=4; ServerHourTo1=22; MAPeriod1=25; MAOffset1=58; ...

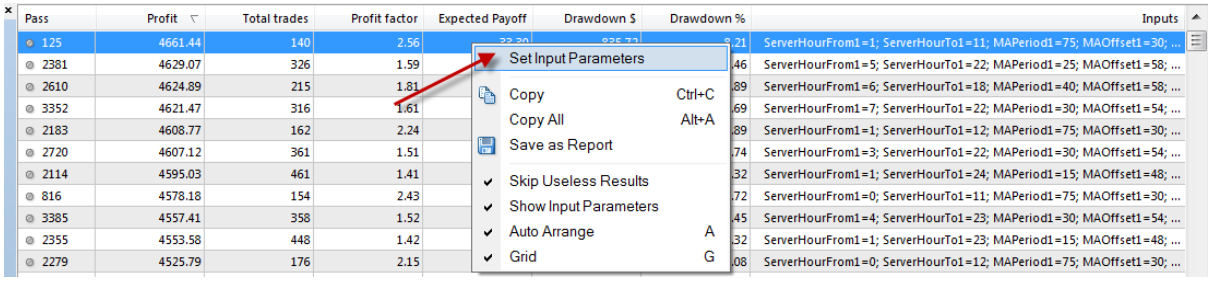
The 125 Pass is nicely placed in a green environment using the MA period and offset settings



The 125 Pass is nicely placed in a green environment using the time of day settings



So the next step is to highlight the option we want to use and to right click on it. You would then click on "Set Input Parameters" and the settings will automatically be loaded into your EA.



Variable	Value
TW1Help	First trading window in b
☑ ServerHourFrom1	1
☐ ServerMinFrom1	0
☑ ServerHourTo1	11
☐ ServerMinTo1	0
MAP1Help	Moving Average Period
☑ MAPeriod1	75
☑ MAOffset1	30

By running a test on these settings the following results are generated

Bars in test	12851	Ticks modelled	24702	Modelling quality	n/a
Mismatched charts errors	0				
Initial deposit	10000.00				
Total net profit	4679.30	Gross profit	7649.58	Gross loss	-2970.28
Profit factor	2.58	Expected payoff	33.42		
Absolute drawdown	654.86	Maximal drawdown	831.44 (8.17%)	Relative drawdown	8.17% (831.44)
Total trades	140	Short positions (won %)	65 (30.77%)	Long positions (won %)	75 (26.67%)
		Profit trades (% of total)	40 (28.57%)	Loss trades (% of total)	100 (71.43%)
	Largest	profit trade	635.57	loss trade	-107.55
	Average	profit trade	191.24	loss trade	-29.70
	Maximum	consecutive wins (profit in money)	4 (1298.68)	consecutive losses (loss in money)	8 (-342.12)
	Maximal	consecutive profit (count of wins)	1298.68 (4)	consecutive loss (count of losses)	-342.12 (8)
	Average	consecutive wins	1	consecutive losses	3

The profit figure is close enough to the expected at $4 661. The profit factor is extremely good. There is a 28% success rate on 140 transactions. We are making a profit because we gain $191 on successful trades and only lose $29.70 on unsuccessful trades. The drawdown % is acceptable.

So as you can see from the above analysis we can find realistic settings that we can apply to live trading in the future for the EURJPY using the 60 minute chart. They are not always the optimal ones as the optimal ones may contain more risk than they should, as we have seen.

Hopefully this process has increased your **confidence and belief** that the Magic Moving Average system can make you money with low anticipated risk levels. You are in a much better position than if you have had to rely on default settings for the rest of the life of this EA.

Optimising other currencies, timeframes and other variables.

You have just learnt a process of optimising the Magic Moving Average Expert Advisor using the EURJPY on 60 min chart. There are hundreds of combinations we have not tested. There must be 18 other currencies that can be tested and a number of timeframes for each currency.

This is a big job. Guide 5 goes quite a long way to helping with this task.

Conclusion of guide 4

I hope that you will agree with me that we have met our objectives for this Guide

This Fourth Guide will cover

- General optimisation considerations
- Optimisation using many variables
- Refining the results using Heat maps
- Loading your settings into the Expert Advisor

Questions and answers

Question 1: Day of the week setup

I am looking at the EURUSD. The settings are different for each day of the week. I am running this on the EU 15 min chart. Also shown is an Average setting. At the moment I have set the EA to the Average settings.

However, to truly use this correctly should I be running a separate EU chart for each day of the week? So I would have on one account on one MT4 platform 5 EU 15 min charts open. One for each day of the week with a separate magic number and each one would be set to trade only on the day of the week assigned for it. Is that what you are recommending??

Answer 1: Day of the week setup

Yes that is exactly what you should do if you want to trade unique settings for each day of the week. The magic number will help you to trace your transactions on the trading account. 5 open charts running separate days of the week should cover this requirement and fully automate the process.

Question 2 - Magic Numbers EA setting

In the manual, you did not make clear if there's a need to change the Magic Numbers if we attach 5 diff charts for EURJPY for each day of the week. I would think it's necessary to change the Magic Numbers for each chart so that each trade for each day is managed independently

Answer 2: Magic numbers EA setting

Yes it looks like you have answered your own question. If you want to manage or monitor the daily transactions in your account give each day a different magic number. If you don't want to be able to identify your results based on their origin, use 1 magic number.

Question 3: Range used for optimisation

For a volatile pair, such as EURJPY, it seems that the criteria of a Moving Average period smaller than 30 is too restrictive? As you can see from my tests I get better results optimising higher ranges.

Answer 3: Range used for optimisation

1) You have full control over the settings you use to do optimise your EA. If you find the results are often at the boundaries of your setting you need to increase the boundaries until you no longer are encountering results that are on the boundaries of you testing range.

2) At the same time you need to apply **realism, common sense and practicality** to your settings. You need to ask yourself if you are really going to be comfortable with 120 MA periods or not. Many people get so carried away with optimisation that they totally lose track of reality. If not why optimise for that range? If you are more comfortable with 30 period Moving Average settings **optimise and test for that range**

Conclusion of this guide

During this course we have used good test data as the basis for implementing the EA, doing the Strategy Tester setup, testing the EA and optimising the settings.

You should now be in a good position to start using the Magic Moving Average Expert Advisor with a high degree of confidence and ownership